MINGEUN CHOI

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EDUCATION

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Mechanical Engineering (Advisor: Dr. Satish Kumar) — Jan. 2023–Dec. 2027 (expected)

Seoul National University — Seoul, Republic of Korea
M.S. in Mechanical Engineering (Advisor: Dr. Seung Jin Song) — Mar. 2017–Feb. 2019

Korea University — Seoul, Republic of Korea
B.S. in Mechanical Engineering — Mar. 2013–Feb. 2017

RESEARCH EXPERIENCE (SELECTED)

Graduate Research Assistant, Georgia Institute of Technology — Jan. 2023–Present

Collaborating with an interdisciplinary team of material scientists and electrical engineers to explore a self-consistent multiscale workflow bridging ballistic-to-continuum physics and device-to-package engineering, achieving the following:

- Designed an automated electrothermal modeling framework to co-optimize vertical power delivery (VPD) architectures and substrate-embedded microfluidic cooling for thermally sustainable, high-power-density operations.
- Developed deep operator network (DeepONet)-based surrogate models for FinFET arrays and VPD packages to rapidly predict spatiotemporal temperature profiles for thermally-aware early design exploration.
- Developing a physics-informed neural network (PINN)-based surrogate model for frequency-domain thermoreflectance (FDTR) to estimate depth-dependent thermal properties of thin films for high-throughput, accurate characterizations.

TECHNICAL SKILLS

- **Engineering and Simulation:** Ansys Icepak (with PyAEDT); Ansys Fluent (with PyFluent); SolidWorks.
- **Programming and Automation:** Python; MATLAB; Julia; LabVIEW.
- **Experimentation and Instrumentation:** Frequency-domain thermoreflectance (FDTR); thermoreflectance thermal imaging (TTI); thermofluidic measurement; machining; manufacturing; commissioning; calibration; uncertainty analysis.

HONORS AND AWARDS (SELECTED)

Technology Commercialization Effort Topping Fellowship, Georgia Institute of Technology — 2026
Excellent Honor Scholarship, Embassy of the Republic of Korea in the USA — 2026
EPS Graduate Student Grant, IEEE Electronics Packaging Society (EPS) — 2026
Professor Avram Bar-Cohen Best Paper Award, Institute of Electrical and Electronics Engineers — 2026
Perspective/Roadmap Paper Topping Fellowship, Georgia Institute of Technology — 2026
Most First Author Journal Papers Topping Fellowship, Georgia Institute of Technology — 2025
Best Paper Runner-Up Award, Institute of Electrical and Electronics Engineers — 2025
Excellence in Service and Leadership Award, Georgia Institute of Technology — 2025
Community Builder/Leader Topping Fellowship, Georgia Institute of Technology — 2025
Strong Gap Funding Topping Fellowship, Georgia Institute of Technology — 2024
Best Presentation Award, Korean Society for Fluid Machinery — 2018
Merit-Based Scholarship, Seoul National University — 2018
Brain Korea 21 Plus Fellowship, National Research Foundation of Korea — 2017
National Science and Engineering Scholarship, Korea Student Aid Foundation — 2015–2016
Academic Excellence Scholarship for Freshman, Korea University — 2013

LEADERSHIP AND SERVICE (SELECTED)

Director of Communications, Georgia Tech Nu Chapter of Pi Tau Sigma — Jan. 2025–Dec. 2025

- Launched a monthly newsletter and redesigned the chapter website, driving a 100 % increase in new-initiate turnout in Spring 2025 and boosting alumni engagement by 30 %.

External Vice President, Mechanical Engineering Graduate Association (MEGA) — May 2024–Apr. 2025

- Coordinated eight professional networking events with Dow Inc., Sandia National Laboratories (SNL), Exponent, and Kratos Defense, increasing event offerings by 50 % and student attendance by 40 % over the academic year.

PUBLICATIONS (SELECTED)

Refereed Journal Articles

[11] P.-C. Lee, **M. Choi**, et al., "Quasi-Ballistic Thermal Transport in Vertically Aligned h-BN," *ACS Omega*, 2026.

[10] S. Krishnakumar, Y. Popryho, **M. Choi**, et al., "A Comprehensive Design Framework for Vertical Power Delivery in High-Performance Computing," *IEEE Trans. Compon., Packag., Manuf. Technol.*, 2026.

[9] **M. Choi** et al., "Deep Operator Network-Accelerated Spatiotemporal Temperature Field Assessment of Fin Field-Effect Transistor Arrays," *Appl. Therm. Eng.*, 2026.

[8] **M. Choi** et al., "Self-Consistent Electrothermal Modeling of Distributed Vertical Power Delivery Architecture with Substrate-Embedded Microfluidic Cooling," *IEEE Trans. Compon., Packag., Manuf. Technol.*, 2026.

[7] D. Go, V. Ashita, A. Wang, **M. Choi**, et al., "Chiplet Encapsulation and Planarization with 100 μm Deep Gaps using AlN Powder Spray + Sputter Hybrid Deposition," *Adv. Eng. Mater.*, 2026.

[6] S. Yun, D. Go, **M. Choi**, et al., "High Speed, High Thermal-Conductivity of Aluminum Nitride Deposited by DC Reactive Sputtering at Low Temperature in the Transition Regime," *Ceram. Int.*, 2026.

[5] P.-C. Lee, **M. Choi**, et al., "High-Speed AlN Film Deposition via Low-Pressure Bipolar High Power Impulsed Magnetron Sputter for Enhanced Thermal Conductivity," *Thin Solid Films*, 2025.

[4] **M. Choi** et al., "Machine Learning-Enabled Fast and Accurate Inversion of Thermal Properties at the Micro- and Nanoscale via Optical Metrology," *Annu. Rev. Heat Transfer*, 2025.

[3] **M. Choi** et al., "Substrate-Embedded Microfluidic Cooling of Distributed Vertical Power Delivery Architectures for High-Performance Computing Processors," *IEEE Trans. Compon., Packag., Manuf. Technol.*, 2025.

[2] P.-C. Lee, A. J. McLeod, **M. Choi**, et al., "Achieving a High Thermally Conductive One Micron AlN Deposition by High Power Impulse Magnetron Sputtering Plus Kick," *ACS Appl. Mater. Interfaces*, 2024.

[1] **M. Choi** et al., "Swirl Enhancement Effect on Turbine Rim Seal Performance," *J. Turbomach.*, 2024.

Refereed Conference Proceedings

[8] **Best Paper** **M. Choi** et al., "DeepONet-Enabled Fast and Scenario-Aware Spatial Temperature Prediction of a Vertical Power Delivery Module for Hyperscale Data Centers," in *Proc. 25th IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, 2026.

[7] R. Dutta, M. Shon, S. Lim, P. Saha, **M. Choi**, et al., "Self-Heating in GAA Nanosheet FETs with integrated Back Side Power Delivery Network: Device to Circuit Electro-Thermal Analysis and Implications," in *Proc. IEEE Int. Reliab. Phys. Symp. (IRPS)*, 2026.

[6] **Best Paper** **M. Choi** et al., "Automated Electro-Thermal Modeling Framework of Distributed Vertical Power Delivery Architectures with Substrate-Embedded Microfluidic Cooling," in *Proc. 24th IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, 2025.

[5] **M. Choi** et al., "Fast Prediction of Spatio-Temporal Temperature Profiles in FinFET Arrays via Numerical and Machine-Learning Approaches," in *Proc. IEEE Int. Electron Devices Meeting (IEDM)*, 2024.

[4] Y. Im, J. Kim, **M. Choi**, et al., "Parametric Thermal Design for Heterogeneously Integrated High-Power Packages," in *Proc. ASME 2024 Int. Tech. Conf. Exhib. Packaging Integr. Electron. Photon. Microsystems (InterPACK)*, 2024.

[3] S. Krishnakumar, **M. Choi**, et al., "Vertical Power Delivery for High Performance Computing Systems with Buck-Derived Regulators," in *Proc. IEEE 74th Electron. Compon. Technol. Conf. (ECTC)*, 2024.

[2] **M. Choi** et al., "Thermal Analysis of High Current Vertical Power Delivery Network with Embedded Microchannel Cooling," in *Proc. 23rd IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, 2024.

[1] S. Krishnakumar, **M. Choi**, et al., "Design Considerations for DC-DC Voltage Regulators in Distributed Vertical Power Delivery Systems," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Singapore, May 2024.

Patents

[3] A. C. Kummel, P.-C. Lee, D. Contreras Mora, D. Go, S. Kumar, and **M. Choi**, "Aluminum Nitride Thin Film Deposition Using Sputtering," U.S. Patent Application No. 19/366,019, 2026.

[2] **M. Choi** and S. J. Song, "Gas Turbine Engine with Improved Flow Stability and Cooling Efficiency," Republic of Korea Patent 10-2206447, filed on Oct. 8, 2020, issued on Jan. 18, 2021.

[1] **M. Choi** et al., "Gas Turbine Engine with Improved Sealing and Operability," Republic of Korea Patent 10-2202368, filed on Sep. 16, 2020, issued on Jan. 7, 2021.